

Invertebrate Phyla

- 3.1 Phylum: Porifera
- 3.2 Phylum: Cnidaria
(Coelenterata)
- 3.3 Phylum: Ctenophora
- 3.4 Phylum: Platyhelminthes
- 3.5 Phylum: Nematoda
- 3.6 Phylum: Annelida
(Earthworm – Type study)
- 3.7 Phylum: Arthropoda
- 3.8 Phylum: Mollusca
- 3.9 Phylum: Echinodermata
- 3.10 Phylum: Hemichordata



3.1 PHYLUM-PORIFERA

(Subkingdom-Parazoa)

- An Evolutionary Blind Offshoot

3.1.1 General Characters

The term *Porifera* (**pore bearing**) was coined and their true ‘animal nature’ was established by **Robert Grant**. Members of this phylum are commonly called sponges. They are generally marine (with the exception of families Potamolepidae and Spongillidae, which include freshwater **sponges** e.g. *Spongilla*). They are either **radially symmetrical** or **asymmetrical** animals (Figure 3.1). These are primitive, **multicellular** and sessile animals and have **cellular level of organisation**. The body wall is composed of two layers (outer epidermis/ pinacoderm and inner choanoderm) separated by a gelatinous matrix called **mesohyl**. Sponges have a **canal system** that constantly conducts water. Water enters through minute pores called **ostia** (inhalent apertures) in the body wall into a central cavity, the **spongocoel** (paragastric cavity), from where it goes out through the **osculum** (exhalent aperture). This transport of water is helpful in gathering food (nutrition-**filter feeders**), respiratory exchange of gases (respiration) and removal of wastes (excretion). **Choanocytes** or **collar cells** line the spongocoel and the canals. Nutrition is holozoic and digestion is ‘**intracellular**’. Reserve food is stored in cells called ‘**thesocytes**’. Nerve cells and sensory cells are absent and so they do not show much co-ordination between the functioning of various parts of the body. The body is supported by a skeleton made up of **calcareous or siliceous spicules or spongin fibres** or both. Sexes are not separate (hermaphroditic or monoecious), i.e., egg cells and sperms are produced by the same individual. Sponges reproduce asexually by fragmentation, budding etc., and sexually by formation of gametes.